



Survey paper

Cyber-security and reinforcement learning — A brief survey

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ABSTRACT

This paper presents a comprehensive literature review on Reinforcement Learning (RL) techniques used in Intrusion Detection Systems (IDS), Intrusion Prevention Systems (IPS), Internet of Things (IoT) and Identity and Access Management (IAM). This study reviews scientific documents such as journals and articles, from 2010 to 2021, extracted from the Science Direct, ACM, IEEEExplore, and Springer database. Most of the research articles published in 2020 and 2021, for cybersecurity and RL are for IDS classifiers and resource optimization in IoTs. Some datasets used for training RL-based IDS algorithms are NSL-KDD, CICIDS, and AWID. There are few datasets and publications for IAM. The few that exist focus on the physical layer authentication. The current state of the art lacks standard evaluation criteria, however, we have identified parameters like detection rate, precision, and accuracy which can be used to compare the algorithms employing RL. This paper is suitable for new researchers, students, and beginners in the field of RL who want to learn about the field and identify problem areas.

1. Introduction

Cybersecurity related problems are a major concern in IoT devices (Tawalbeh et al., 2020). Since many of these devices are constantly moving between different networks it makes them vulnerable to risks. New environmental laws and privacy protection regulations, like the GDPR (Anon, 2018) have increased the processing overhead of existing solutions. Therefore, accurate and reliable Machine Learning-based (ML) malware detection and prevention systems are essential (Xiao et al., 2017).

RL is a strong and capable ML technique (Sutton and Barto, 2018). In this literature review paper, we present a unique combination of reinforcement learning, IoT, and cybersecurity. Currently, several surveys are related to RL, however, they do not present the applications of RL in IDS-IPS, IoT, and IAM systems. Review papers like Xiao et al. (2018b) present RL applications in network security. Several surveys present ML for networking (Wang et al., 2018a) too. However, they focus on Deep Learning (DL) approaches. We have summarized the existing survey papers on this topic in Table 1.

We can see from Table 1, that no survey presented RL methods for cybersecurity in the domain of IoT and IAM. This inspired us to conduct a comprehensive literature review and survey on the applications of RL to address issues in IDS-IPS, IoT, and IAM systems. This paper is ideal for novice researchers, students, and beginners in the domain of RL to explore and identify research areas.

Readers can refer to Fig. 1 for a quick overview of the evolution of research in RL-based methods, algorithms, and applications in literature extracted from IEEEExplore. A pronounced observation is that RL for identity management has not been investigated. Therefore, the potential for research in this domain is immense.

This paper is structured as follows. In Section 2, RL concepts. Section 3 describes the survey methodology followed by a literature review in Section 4. Section 4 is divided into three RL-based sub-sections. In Section 5, we present existing issues and their future scope. After presenting a brief conclusion in Section 6, we close with references. This is the first comprehensive and visual survey available online that covers the RL algorithms in the three domains of IDS/IPS, IAM, and IoT.

2. Reinforcement learning: an overview

In this Section, we discuss the core concepts of Markov Decision Process (MDP), theory of RL, ϵ -greedy policy, Q-learning, DRL, and the metrics for evaluating RL algorithms.

2.1. Markov decision process - MDP

A Markov decision process is a controlled stochastic process of representing and solving problems where there is uncertainty and sequential decision determines the result. To describe a typical MDP

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Table 1
Summary of existing survey papers.

Study	Summary Points	Comments/Novelty
Wu et al. (2021)	The authors evaluate blockchain and ML applications in Industrial IoT (IIoT) from three key perspectives: consensus mechanism, storage, and communication. The future scope given in this survey outlines the unresolved problems that must be addressed before large-scale deployments of IIoT applications are feasible.	We review the blockchain and RL-based applications in IoT, IDS, IPS and IAM application domains (Tao et al., 2021; Ali and Alam, 2019; Hao et al., 2021).
Nguyen and Reddi (2021)	This paper discusses Deep Reinforcement Learning (DRL) based security methods for cyber-physical systems, autonomous intrusion detection techniques, and multiagent DRL-based game theory simulations for cyberattack defensive strategies.	The research methodology of this paper and Nguyen and Reddi (2021) is different. We approach the reinforcement learning and cybersecurity topic from the angle of three major application areas. Also, this paper has a broader scope since literature other than 'Deep' Reinforcement Learning is surveyed.
Nugroho et al. (2020)	An excellent paper that discusses all three categories ML Algorithms Supervised learning, Unsupervised Learning, and Reinforcement Learning for IDS IoT.	With respect to this paper, our paper is novel in terms of having a scope limited to the Reinforcement Learning algorithms for IDS used in multiple applications including IoT.
Tawalbeh et al. (2020)	The authors explore the nuances of security and privacy of IoT systems. Also, a novel IoT layered model with detailed privacy and security components is proposed.	Although the cybersecurity challenges in IoT systems are addressed in Tawalbeh et al. (2020), ML-based solutions are not mentioned in the paper.
Jagannath et al. (2019)	This work begins with an overview of all three categories of ML-based techniques, at each layer of the OSI model, to solve the problems in IoT wireless communications and networks. Discussions about the feasibility of the hardware implementations. Each section has a detailed analysis of the future scope.	We present a comprehensive survey of RL-based methods only for multiple application areas.
Wang et al. (2018a)	The selective survey performed in Wang et al. (2018a), pivots around the workflow of ML in the networking domain. The applications vary from resource management to traffic prediction and classification.	The application areas of RL are identified in our work. RL has several applications in resource management and other areas of the networking domain.
Arulkumaran et al. (2017)	The objective of this survey is to cover the DRL algorithms like Deep Q-network (DQN), trust region policy optimization, and Asynchronous Advantage Actor-Critic (A3C).	There is no mention of the application areas of IoT and IAM in the current areas of research of Arulkumaran et al. (2017).

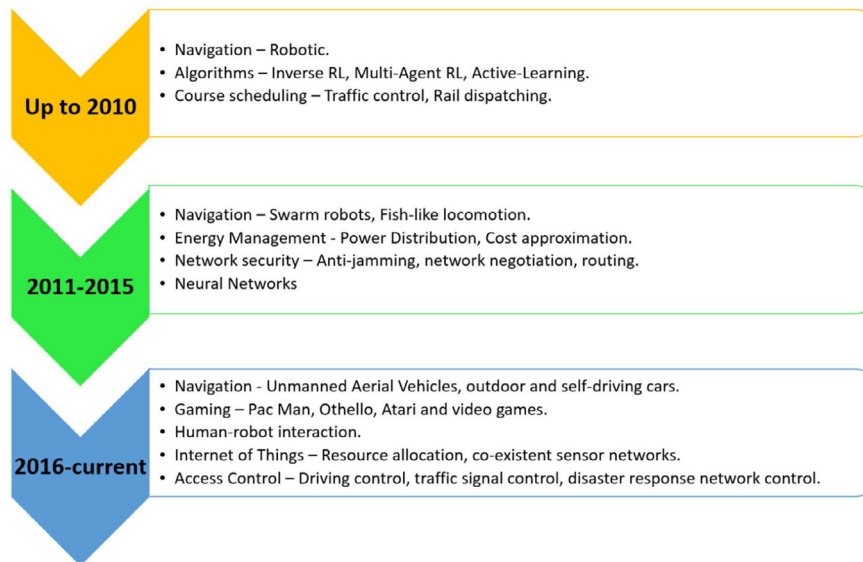


Fig. 1. Evolution of the Reinforcement Learning literature in IEEE Xplore.

problem (Sigaud and Buffet, 2013), the following five-element tuple is used (S, A, T, p, r):

- S — the state space;
- A — the set of all possible actions;
- T — the set of time steps where decisions need to be made;
- pO — the state transition probability function;
- rO — the reward function defined on state transitions.

For each time step t in T , the agent is in a current state S_t . An action A_t chosen in this current state affects the transition to the next time step $t+1$ and next state S_{t+1} .

Reward r_t is obtained by the agent for this transition from S_t to S_{t+1} . The reward r_t can be received at the immediate time t or accumulated over the period t and $t+1$. The same is illustrated in Fig. 2. This process continues iteratively until 'end condition' is met.

2.2. Reinforcement learning

RL is the paradigm of ML where a decision-making agent interacts actively with its uncertain. It is also called an extension of dynamic programming (Bellman, 1952) and MDP. The problems where the system's actions influence its later inputs are known as closed-loop problems, and RL is the solution for such problems. It is different from MDP in the sense that the transition and reward functions are not known beforehand. The most important components of the RL system are the following:

- Agent — The primary component of RL is a goal-seeking agent that is highly interactive with its uncertain environment.

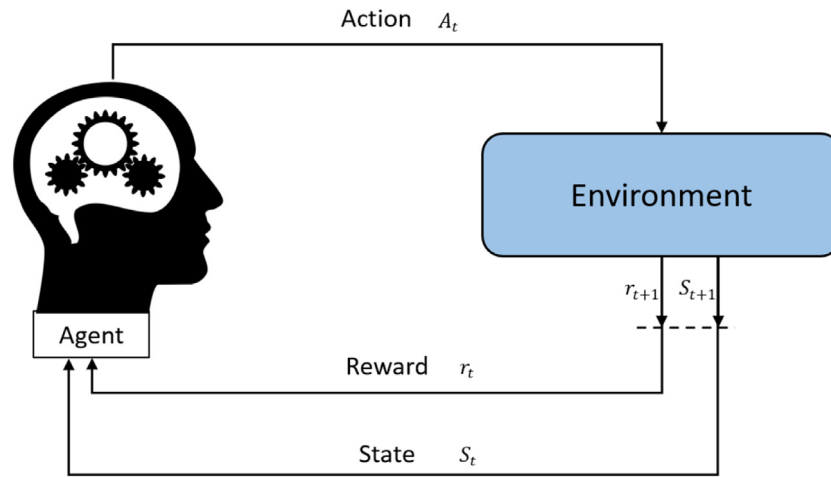


Fig. 2. Working of MDP.

- Environment — Everything the agent interacts with is called the environment. The environment has certain features and attributes. The agent senses and evaluates these attributes in each state. The agent uses this information to choose actions that in turn could influence the outcomes.
- Policy — A map between the states in the environment and actions that can be taken from that state. It can be a lookup matrix, a function, or a search function.
- Reward signal — The reward signal defines the goal of the problem. At each time step, the environment assesses the current state and action of the agent and sends a single value to the agent. This value is the reward. It can be negative or positive. The sole aim of the agent is to maximize this reward.
- Value function — It is defined for a state. When the agent starts from this state and accumulates a total reward over the future, it is known as the value of the state. The estimate the values, iterations are run over the agents' sequences of observations.

RL methods are classified into model-based and model-free methods based on this concept. Model-free methods lack planning and information about the previous actions-states. These are trial and error-based methods.

2.3. Exploration vs exploitation

When an agent discovers an action from one state to another, which has a high reward and tries performing the same action over and over to increase its cumulative reward, the agent *exploits* its current knowledge. Also, the action itself is called *greedy*. If the agent chooses an action from one state about which it has some knowledge to another state about which it has little or no knowledge, the agent is *exploring*.

To further explain the exploration and exploitation, we will take the example of a mouse acting as an agent in a 4×4 environment grid or matrix as seen in Fig. 3. Here, the rewards are the preferable states and are shown by cheese slices. The epoch ends if the mouse ends up in the same state as the cat. The mouse may identify the path $21 > 11 > 12 > 22 > 21$ as a path with high rewards. Once it identifies the actions required to move into these states, it may repetitively move only in these states. This would be the exploitation of the knowledge acquired. However, if the mouse ventures beyond the known path and moves from $42 > 43$ to 44 . It can discover an even larger reward! This is exploration.

As He Fang, et al. mention in Fang et al. (2019), RL pivots on the equilibrium between the exploration of the new versus the exploitation of the known. When problems require decision-making, the use of DL amplifies the RL capabilities in deciding (Arulkumaran et al., 2017).

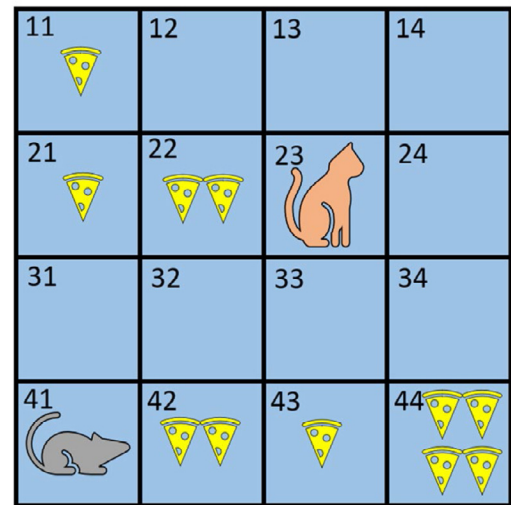


Fig. 3. Exploration vs Exploitation.

This became evident with the Deep Reinforcement Learning (DRL) algorithm in Mnih et al. (2016) that played Atari 2600 video games at extraordinary speed and efficiency.

2.4. Q-learning

The most common RL algorithm is Q-learning. Updates are a function of the optimal actions notwithstanding the actions performed. Knowledge of the action in the next step is not required.

When real experiences of Q-learning are combined with the simulated experiences of the model to reach convergence faster, the algorithm is known as Dyna-Q.

2.5. Deep reinforcement learning

It is clear from the above Sections that a clear definition of state and action is important to ensure appropriate agent training. Let us consider an IDS for an Aerial Computing Network (Tao et al., 2021) or a Cloud Infrastructure (Sethi et al., 2020). The state and action space in both these cases is high-dimensional. DL uses artificial neural networks to approximate the policy and value functions. It eliminates the need to use a lookup table to map states and actions to values. The combination of DL and RL, albeit DRL algorithms use samples from the state or action space to reach the end goal.

Algorithm 2 The Q-learning Algorithm

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Initialize  $Q(s, a)$ ,  $\forall s \in S, a \in A(s)$ , arbitrarily, and  $Q(\text{terminal-state}, \cdot) = 0$ 
Repeat (for each episode):
  Initialize  $S$ 
  Repeat (for each step of episode):
    Choose  $A$  from  $S$  using policy derived from  $Q$  (e.g.,  $\epsilon$ -greedy)
    Take action  $A$ , observe  $R$ ,  $S'Q(S, A) \leftarrow Q(S, A) + \alpha[R + \gamma \max_a Q(S', a) - Q(S, A)]$ 
     $S \leftarrow S'$ ;
  until  $S$  is terminal

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In this survey paper, we review several research articles that use DRL in different applications. Some of the popular DRL algorithms are DQN (Wang et al., 2019), Deep Deterministic Policy Gradient (DDPG) algorithm (Tao et al., 2021) and Double Deep Q Network (DDQN) (Caminero et al., 2019).

2.6. Metrics of evaluation of reinforcement learning

For qualitative comparisons of RL algorithms in the domains we have considered, there are three main metrics of evaluation

1. Accuracy — Authentication accuracy is a critical performance criterion. The authentication accuracy is measured using the metric values mentioned below.
 - a. False Acceptance Rate (FAR): The FAR measures the likelihood of accepting an invalid user.
 - b. Missed Detection Rate (MDR): MDR reflects the likelihood of failing to detect an invalid user.
2. Computing Efficiency — Time complexity and time cost are major indicators of system performance. Taking less time implies being more efficient. In an earlier Section, we come across the term ‘greedy’. The term ‘greedy’ relates to computing efficiency of the RL algorithm. The meaning of greedy action selection is that the algorithm (usually) selects the next action with the best reward. If an immediate reward is considered, it can be short-sighted learning algorithm. It is preferable to have the greedy action selection such that a possibly longer and better path is chosen. The computing efficiency also comprises the storage space efficiency in several applications. Considering many IoT applications that are mainly a resource-constrained environment.
3. Time of Convergence — Above we discussed time as a part of computing efficiency. Here, we discuss a specific time of convergence of the RL algorithm. When an RL agent is first introduced into the environment, it is likely to perform thousands if not millions of actions and experiments to reach optimal learning. It is ideal to have a model in place which can assist in reducing the number of steps and interactions required with the environment, thereby reducing the time to convergence.

Several quantitative metrics are available that can help standardize the comparison of RL algorithms (Agarwal et al., 2021).

3. Research and survey methodology

A mixed research methodology was adopted. We begin with exploratory research by reviewing the domains of ‘reinforcement learning’ and ‘cybersecurity’. After identifying the challenges, the selected

database is searched using keywords, followed by qualitative evaluation of the results. We summarize and compare the relevant papers from the search results. Finally, critical comments are provided for the discussion groups IDS, IPS, IoT and IAM within the scope of RL. A systematic literature review is illustrated in Fig. 4.

Our main motivation is to identify the solutions that use RL to solve cybersecurity challenges, with the aim of answering the following Research Questions (RQ):

[RQ1]. How many papers use RL:

- a. To build Intrusion Detection and Prevention Systems?
- b. For Identity and Access Management?
- c. For various aspects of IoT devices and networks?

[RQ2]. What is the current state-of-the-art RL based methods for IDS, IPS, IoT and IAM?

[RQ3]. What are the elements of the RL algorithm used in the state-of-the-art papers?

[RQ4]. What datasets can be used in IDS-IPS models with an RL approach?

[RQ5]. What is the future scope of using RL to solve the challenges of cybersecurity and IoT?

We attempt to answer some of these RQs in this Section through the following survey methodology. The search terms used are listed in the description of the figures shown below. The analysis of publications available in Scopus indexed literature is shown in Fig. 5. Data access information source: <https://www.scopus.com> (accessed on 26th November 2021). As we can see, there has been a rapid increase in research done in IoT in the last couple of years, especially in 2021. The literature considered here is all-inclusive of journals, conferences, books, articles and more.

Looking at the graphs in Fig. 6 we can see that there has been a jump in using reinforcement learning in the domain of IDS, and IPS.

Primarily, publications from 2016 to 2021 were included. The publications for keywords ‘reinforcement learning’ and ‘intrusion detection’ in SCOPUS identified 553 works, Springer displayed 1421 studies, while IEEE returned 54 research works. Whereas the keywords ‘reinforcement learning’ and ‘intrusion prevention’, SCOPUS identified 274 works, Springer displayed 1027 studies, while IEEE returned only seven research works. Document publication per year is presented in Fig. 6. Data access information source: <https://www.scopus.com>, <https://ieeexplore.ieee.org/>, and <https://link.springer.com/> (accessed on 26th November 2021). Several studies did not come up in the results of the search query. But they were included in this review. They were identified from the references in relevant papers.

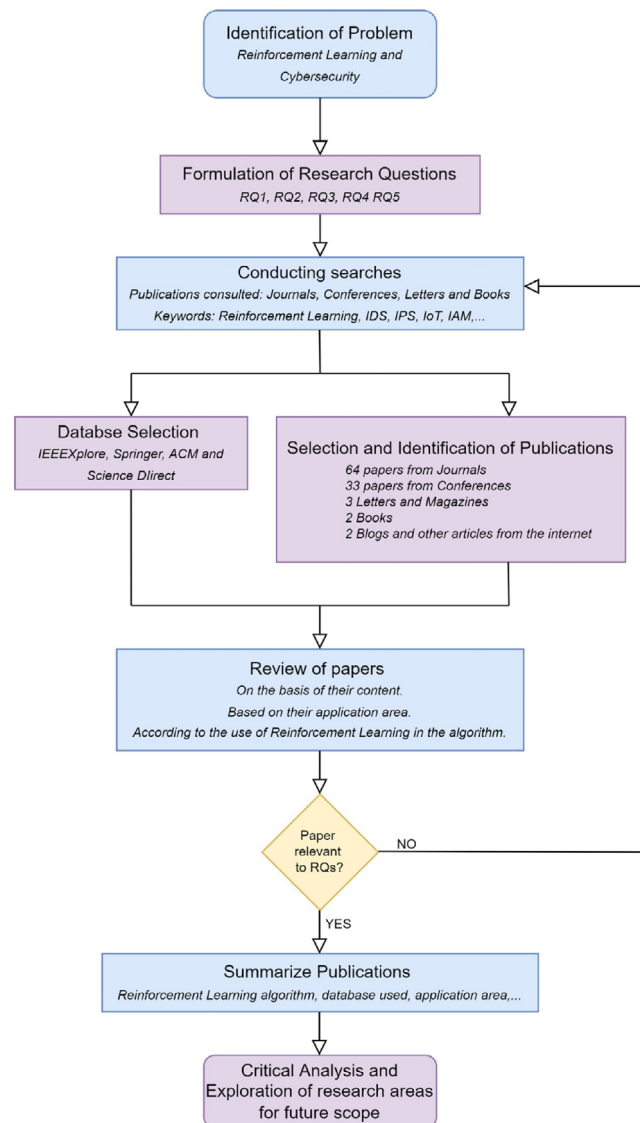


Fig. 4. Research Methodology.

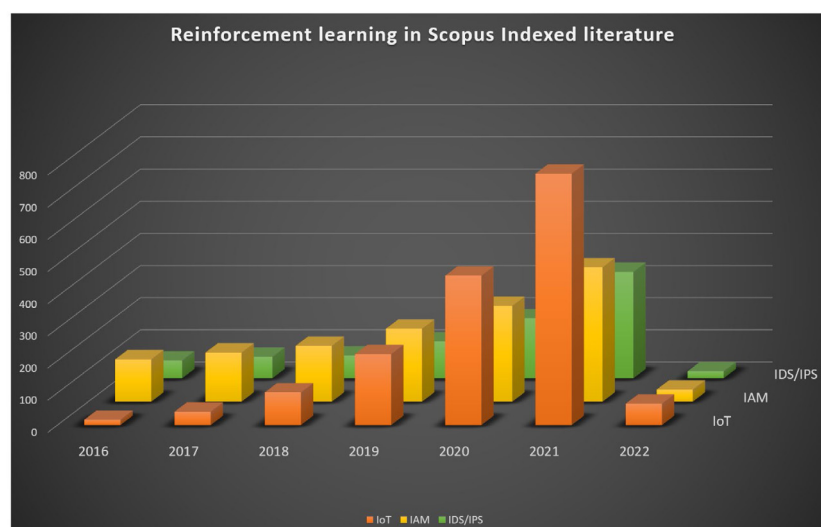


Fig. 5. Year-wise distribution of publications in Scopus indexed literature in IDS/IPS, IoT and IAM for years 2016–2022.

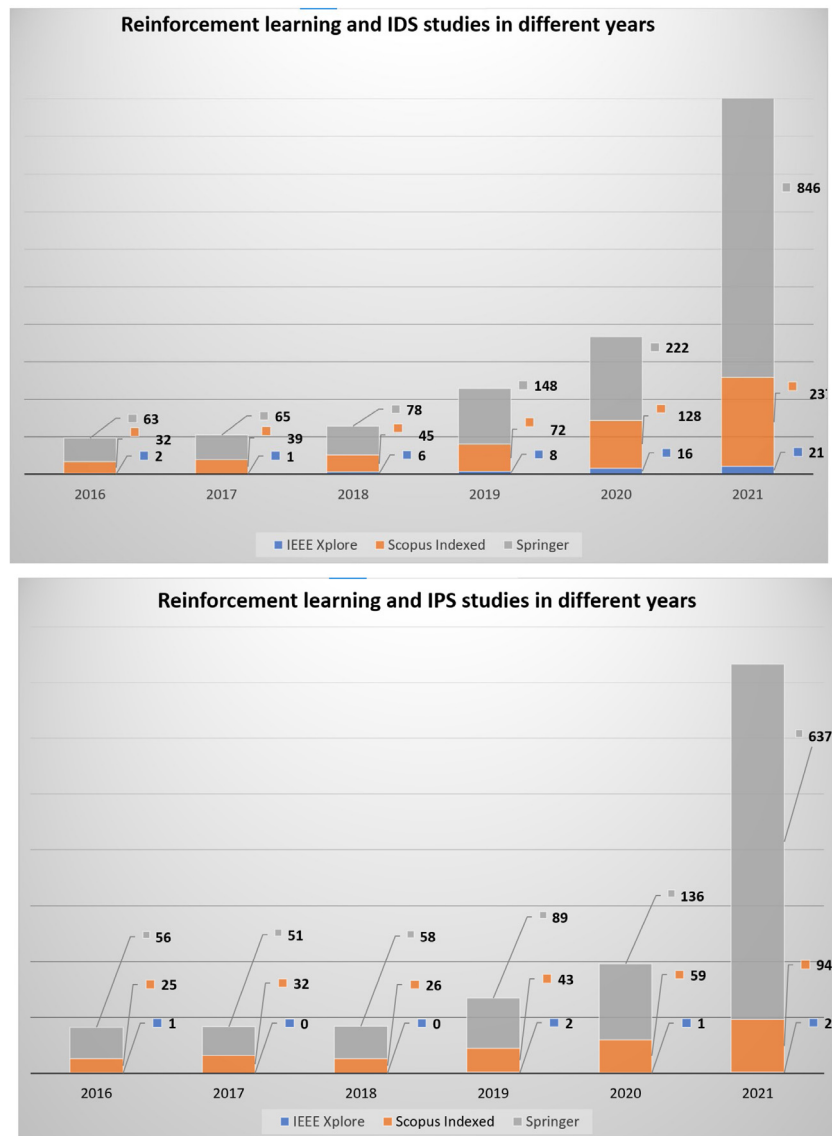


Fig. 6. Reinforcement learning and IDS and IPS studies and years.

The number of publications for keywords ‘*reinforcement learning*’ and ‘*access management*’ are as follows. SCOPUS identified 6574 works, Springer displayed 14748 studies, while IEEE returned 695 research works. Lastly, for the keywords ‘*reinforcement learning*’ and ‘*authentication*’, SCOPUS identified 639 works, Springer displayed 1486 studies, while IEEE returned 36 research works. Data access information source: <https://www.scopus.com>, <https://ieeexplore.ieee.org/>, and <https://link.springer.com/> (accessed on 26th November 2021). Document publication per year is presented in Fig. 7.

Searching for the keywords ‘*reinforcement learning*’ and ‘*IoT*’, SCOPUS identified 1630 works, Springer displayed 3861 studies, while IEEE returned 663 research works. Data access information source: <https://www.scopus.com>, <https://ieeexplore.ieee.org/>, and <https://link.springer.com/> (accessed on 26th November 2021). This is seen in Fig. 8.

4. Applications and uses of reinforcement learning

4.1. Reinforcement learning and IPS and IDS

Most ML-based intrusion detection frameworks are built on the concepts of supervised learning (Marir et al., 2019; Kotani and Sekiya,

2019; Wei et al., 2017; Gao et al., 2018; Seo et al., 2017; Bennaceur et al., 2019; Wang et al., 2018b; Fernandez and Xu, 2019; Rustam and Talita, 2018; Mirza and Cosan, 2018; Pamukov and Poulkov, 2017). In this Section we review several papers that focus on use of RL. The authors of Caminero et al. (2019) present a novel intrusion detection algorithm called ‘Adversarial Environment using Reinforcement Learning (AE-RL)’. This algorithm uses the DDQN algorithm, which is a combination of RL and supervised learning. The two agents are — an environment selector and a classifier. The two agents are adversarial to each other because the rewards of one counter the rewards of the other. NSL-KDD (Tavallaei et al., 2009) and AWID (Kolas et al., 2016) datasets are used for training and testing.

As seen above, the aim is to classify and detect attacks and anomalies. Many papers have the same objective (Sethi et al., 2021, 2020; Tao et al., 2021; Gu et al., 2020; Puzanov et al., 2020; Bouhamed et al., 2021; Suwannalai and Polprasert, 2020; Heartfield et al., 2021; Ma and Shi, 2021; Otoum et al., 2021; Mohanty et al., 2021; Lopez-Martin et al., 2021). MAGPIE is an anomaly classification architecture suggested by Heartfield et al. (2021). It combines real-time unsupervised anomaly detection and a Multi-Armed Bandit reward generation RL algorithm to select models based on human presence inference in a smart home. Another study presents an intrusion detection framework

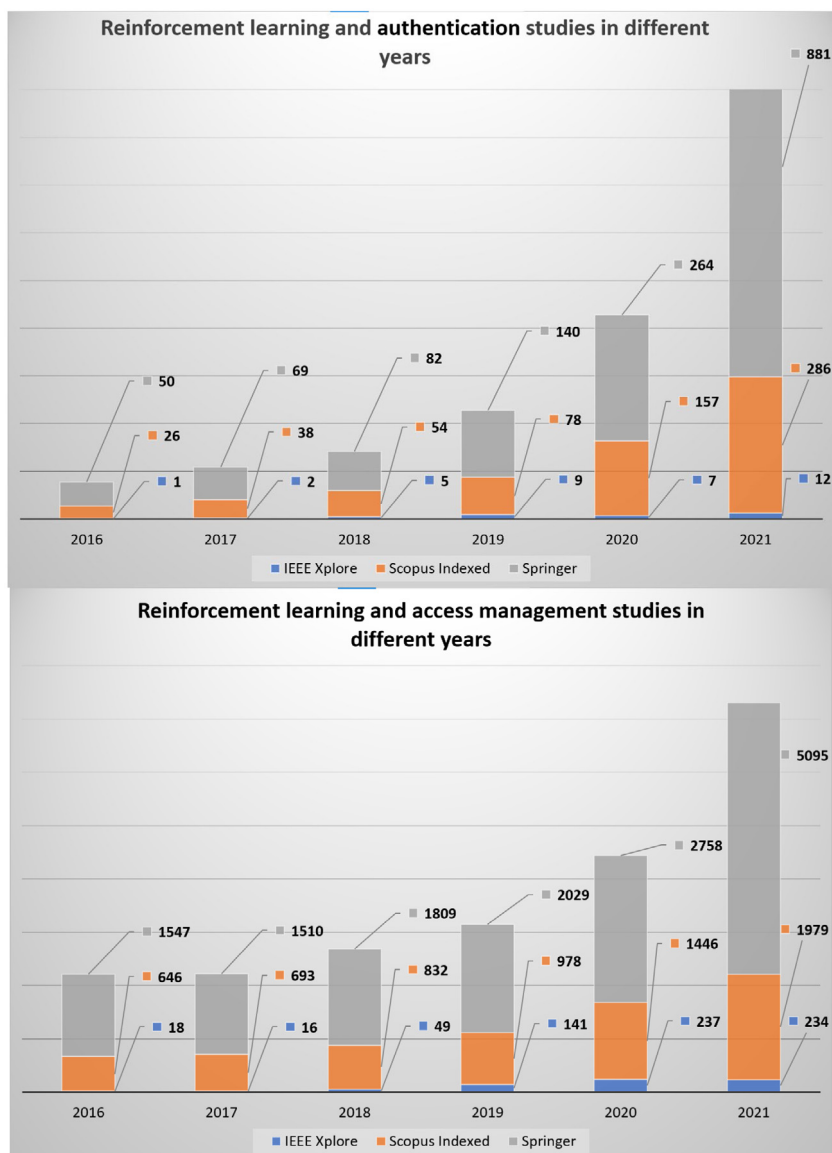


Fig. 7. Reinforcement learning and access management and authentication studies and years.

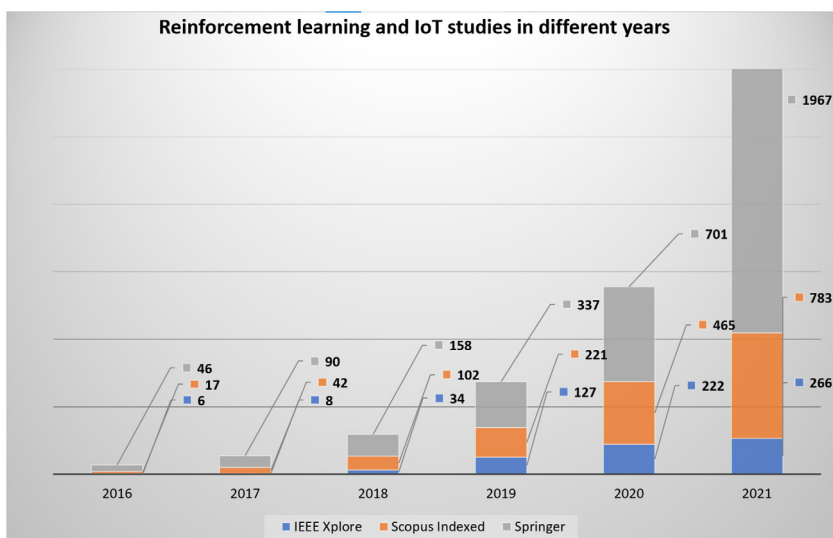


Fig. 8. Reinforcement learning and IoT studies and years.

that combines a primary critic neural network, a primary actor neural network, and Q-learning for intrusion detection and classification activities (Tao et al., 2021). The use of radial basis function neural network for the classification of attacks is shown by Lopez-Martin et al. (2021). This method uses offline DRL to achieve quicker optimization of network parameters.

Papers Tao et al. (2021) and Bouhamed et al. (2021) implement DRL for IDS classifiers in Unmanned Aerial Vehicles (UAVs). The communication technologies used in autonomous vehicles and mobile devices is such that they are vulnerable to several forms of attacks. The technique given in Rasheed et al. (2020) uses LSTM-GAN with a new adversarial DQN, to make autonomous vehicles resilient to the defective sensor data insertion attack by adversaries. This is discussed in more detail in the last Section which deals with IoT. In Xiao et al. (2017), the authors provide an algorithm for malware detection when mobile devices offload application traces to the cloud. The design novelty of the proposed malware detection scheme is that it combines Nash Equilibrium from Game theory (Hu and Wellman, 2004) with a Post-Decision State (PDS) learning-based scheme, Q-learning, and Dyna Architecture of RL.

Penetration testing is a common practice to test the resiliency of IDS classifiers and security of networks in general. Authors of a recent study perform SQL injection to exploit vulnerabilities of webpages and databases using a Deep Neural Network (DNN) to approximate Q-values of the Q-learning algorithm (Erdödi et al., 2021). Use of the neural network makes it easier to deal with unwieldy action and state spaces. Post-attack analysis of a cyber-attack is performed if several vulnerabilities are exploited, and the network is compromised. In Maeda and Mimura (2021), the authors present a novel technique to automate post-attack analysis. This technique uses the Advantage Actor-Critic (A2C) algorithm with a DNN of 5-layers that ensures the experience is accumulated. Many countries have laws which enforce organizations to publish information related to a hack within a stipulated time frame. Therefore, organizations give immense importance to their networks.

The paper by Li et al. (2017), considers a scenario where a communication network is considered. Q-learning helps the first agent, the sensor decides on an ideal transmission power strategy in the presence of the second agent, the attacker evaluates the remote estimation performance (Elderman et al., 2017). Load balancing and computation offloading is another metric used to maintain network performance.

The authors of Aref et al. (2017) used Q-learning to decide on the optimum sub-band selection policy for Wideband Autonomous Cognitive Radios (WACR). The RL algorithm helps the WACRs learn the functioning paradigm of the peer WACRs and the jammers. Jamming prevention, and interference from other radios using the same spectrum is avoided. Eventually, the policy selection decision-making abilities are enhanced. Summaries of the algorithms and models reviewed above are presented in Table 2 along with their future scope. We have identified several datasets used for IDS and IPS applications. This is described in the following sub-section.

4.1.1. Datasets

RL methods can learn from the environment without training on a dataset. However, solutions use DRL methods for IDS/IPS problems. Datasets are required in such cases for training and testing the model. Table 3 summarizes some datasets that we came across during the literature review. In several papers, the authors have opted for self-generated datasets and synthetically created datasets via different ML algorithms (Gu et al., 2020; Venturi et al., 2021; Lu et al., 2020; Ring et al., 2019; Hsu and Matsuoka, 2020). The number of datasets available for IoT is fewer than that available for IDS/IPS. Whereas for IAM, there are hardly any datasets and publications available.

To conclude the IDS and IPS Section of this survey paper, we refer to another survey study (Ling et al., 2015), which focuses on how RL supplements the security strategies for cognitive radio networks. It covers multiple attacks and the RL models that exist to prevent them. This article proposes methods for implementing authentication, authorization, and access control as an additional safeguard against those malicious activities that escape the IDS and IPS RL models. This is elaborated in the next Section.

4.2. Reinforcement learning in identity and access management systems

It is challenging for a gateway to authenticate each device, signal and packet coming into the network and accessing the resources. Therefore, it is imperative to empower these gateways with an intelligent authentication mechanism like RL.

The first paper reviewed here was authored by Ferdowsi and Saad (2019). A novel algorithm is proposed which involves authentication of IoT signals by discovering and identifying the adverse IoT devices. DRL is used to predict which IoT devices will be adverse or vulnerable, thereby assisting the already overburdened gateway to make quicker decisions.

The next two papers authored by Liang Xiao et al. propose an authentication game in Universal Software Radio Peripherals (USRP) using Q-learning, where packet acceptance or rejection depends on the threshold-level θ (Xiao et al., 2015). In Xiao et al. (2016), the authors expand this research and present the same spoofing detection scheme that includes Dyna-Q along with Q-learning to increase authentication speed and accuracy of threshold determination. The authors Ning Gao et al. solved the problem of intelligent spoofing attacks by suggesting a Q-learning-based PHY-layer authentication mechanism to accelerate the cooperative neighbor sensor selection, which in turn helps achieve belief convergence faster in Wireless Sensor Networks (WSN) (Gao et al., 2020).

In this paper Xu et al. (2019), the authors propose another physical authentication mechanism that uses RL on the voltage signal from the Electronic Control Unit of the CAN bus to determine a threshold for authenticating to the other parts of the vehicle like engine control or braking device. This concept was explored in a recent study by Xiao et al. (2021). Thus, the common trend observed is that RL-based authentication mechanisms are implemented, which determine the threshold parameter using Q-learning, Dyna-Q, etc. (Xiao et al., 2015, 2016; Tao et al., 2021; Xiao et al., 2017, 2018b).

We discuss two mobile crowdsensing (MCS) application authentication schemes from article (Xiao et al., 2018a). The first is a DNN-based authentication scheme that regularly monitors the mobile users and uses distinct attributes of Wi-Fi signal. The second authentication method reviewed in Xiao et al. (2018a) uses a 3-hidden-layer Stacked Auto-Encoder (SAE) based authentication with a parametric Rectified Linear Unit (ReLU) activation function. This algorithm identifies and classifies authentic users from the illegitimate ones. Both are unsupervised learning authentication methods and not categorized under RL. However, the best and most secure authentication schemes combine different methods (Wang et al., 2019; Vasilyev et al., 2018; Khan et al., 2017).

Some common targets of cyberattacks are IoT devices like cars and vehicles (Greenberg et al., 2020). The study by Lu et al. (2020) addresses the problem of rogue edge attackers in Vehicular Ad hoc Networks (VANETs). A PHY-layer authentication policy implements the Dyna architecture to create a simulated authentication experience to train itself. Combination with a deep Convolutional Neural Network (CNN) reduces the time required and increases authentication accuracy. Similarly, the authors of Xiao et al. (2019) in the presence of an authentication policy with the Power Delay Profile (PDP) scheme, abbreviated as RAP for Underwater Acoustic Sensor Networks (UWSNs). This policy is improved in accuracy and speed by integrating it with CNN and is known as DRAP. Speed and accuracy can also be improved by using ML.

He Fang et al. present a survey of authentication mechanisms using different ML techniques in Fang et al. (2019). The authors discuss the RL-based PHY-layer authentication mechanism given in Xiao et al. (2016) and suggest combining it with high-level application layer authentication to make a robust and dynamic policy. The high-level application layer authentication is based on the attributes of end-users. These attributes are usually categorized as Sensitive Private Information (SPI). Knirsch et al. (2018) and Iacovazzi and Baiocchi (2014)

Table 2

Summary of reinforcement learning methods used in IDS and IPS.

Reference	Model/ Application	Learning algorithm	Agent	States	Actions	Reward	Future Scope
Caminero et al. (2019)	AE-RL	Q-learning, DDQN	Classifier and Environment	Random samples from NSL-KDD and AWID data	Intrusion label prediction	Value of +1 or 0	(1). Re-apply the trained environment for other classification tasks. (2). Implementation in a prioritized experience replay application.
Tao et al. (2021)	DDPG scheme	DDPG algorithm; Actor–Critic network with replay memory	UAVs	Based on the following behavior specification: Normal behavior of UAV — safe states; abnormal behavior of UAV — malicious states; Limit number of states by identifying the illegitimate values.	Binary value. UAV malicious and detected = 1, Otherwise 0.	Reward is system utility-defined The negative of the system is lost. An attack is conducted but is not detected, leading to the system being lost	(1). Integration with Blockchain. (2). Using 6G networks for UAV. (3). Solar energy-powered UAV.
Bouhamed et al. (2021)	IDPS with periodic offline learning and a customized reward function	DQN, Pearson Correlation for training dataset	UAVs — classifier	A sampling function from the training dataset CICIDS2017	$A = \{0, 1, \dots, n-1\}$, n is the number of investigated classes	Reward is a function of state (majority and minority classes), action and label. Its value range is $+\beta$, $-\beta$, $+\zeta$, $-\zeta$, where 1 $\geq \beta > \zeta > 0$; and C_M and C_m are the majority and minority classes.	The dataset used for training has 70 features. Of the 70 features, only 10 features are used. Future scope should evaluate the accuracy of the model considering this small number of features selected. Also, if the accuracy can be improved by increasing the number of features.
Xiao et al. (2017)	Malware Detection	Q-learning, Dyna-Q, Post-Decision State Learning	Mobile device	Radio bandwidth and the previous offloading rates of the other devices	Offloading rate of the mobile device	Utility and result from server	Improve the performance of PDS-learning detection when estimated channel state transition is inaccurate.
Heartfield et al. (2021)	MAGPIE	Multi-Armed Bandit	Classifier	Meta-data stream snapshots	Determine anomaly detection decision threshold	The silhouette score from dataset clusters represents a measure of cluster similarity	(1). Federated learning (FL) dynamically determines the optimal q value. (2). Proactive defense mechanism against emerging adversarial ML attacks. (3). Build resiliency against detection accuracy disruptors.
Li et al. (2017)	SINR-Based DoS Attack on Remote State Estimation	Markov Game, Q-learning	Sensor and Attacker	Sensor — Successful reception of ACK from a remote estimator. Attacker — successful eavesdropping of ACKs	Changes in the transmission power or interference power	State estimation error covariance and Energy consumption	(1). Improve the estimation process for finding a weaker sufficient or necessary and sufficient condition. (2). Improve resiliency stability system when the attacker has a dominant energy budget.
Aref et al. (2017)	MARL	The partially observable Markov decision process (POMDP)	WACR	Sub-band states 0/1. 1 (available), if the idle bandwidth in the sub-band is greater than or equal to minimum required bandwidth for transmission. Otherwise, 0 (unavailable).	Selecting a sub-band	Function of time taken for the jammer/interference signals to interfere with a transmission	Enable Wideband autonomous cognitive radios (WACRs) to select optimal sub-band in the presence of jammer equipped with cognitive radio technology.

(continued on next page)

present ML techniques to generate obfuscation patterns for masking and unmasking these attributes. An outline of the discussions in this Section is presented in [Table 4](#). We conclude this Section by discussing [Buduru](#)

and [Yau \(2015\)](#). The authors here present an Adaptive Continuous User Authentication technique for mobile users where the MDP policy learns end-user finger gestures. Following this, while installing

Table 2 (continued).

Reference	Model/ Application	Learning algorithm	Agent	States	Actions	Reward	Future Scope
Lopez-Martin et al. (2021)	RBFNN with Offline RL	Q-learning	RBFNN model	Features of samples in the chosen dataset ((NSL-KDD, UNSW-NB15, AWID, CICIDS2017, and CICDDOS2019)	Selection of a possible intrusion label	Discounted maximum Q-value	(1). Make the action space manageable when it becomes too large due to increased dimensionality. (2). Explore new loss functions like contrastive loss to train RBFNN. (3). Experiment with other algorithms like Actor-Critic, policy-gradient, adversarial. (4). Implement eXplainable Artificial Intelligence (XAI) techniques to extract the classifier process.
Erdödi et al. (2021)	SQL injection vulnerability exploitation using Q-learning	Jeopardy-style Capture the Flag (CTF) game, MDP, Q-learning and DQN	Potential attacker/pen tester	History of actions taken, and responses seen	Finite set of SQL strings	Positive — SQL injection was successful. Negative when it is unsuccessful.	(1). Testing with real-world case of SQL injection. (2). Improve a given technique by using a large combinatorial action space. (3). Test with more vulnerabilities like error-based or time-based SQL injection. (4). Test against HTML pages from a web server.
Maeda and Mimura (2021)	DRL with the PowerShell Empire	A2C	PowerShell	10 parameters – for example – compromised computer, admin access, computer found	Selected one of the PowerShell Empire modules	$r = 50$ - > valuable Lateral Movement is successful. $r = 10$ - > low-value Lateral Movement is successful. $r = -1$ - > the lateral movement fails. For other actions, $r =$ -1 regardless of success or failure.	(1). Methods to diversify the learning environment. (2). Designing and evaluating other definitions of the agent's state.

new applications, software updates, and making configuration changes, this algorithm uses the information learned over time to authenticate the user. This proposal points us toward pervasive continuous authentication of IoT devices that have unlimited applications.

4.3. Reinforcement learning in IoT networks

Smart cities are a large complex network of IoTs in which shared computing is implemented to efficiently use resources. The requirement for end-users in smart cities is that responses should be quick, and caching helps achieve this. Caching always consumes the resources of space and memory. Therefore, the major problems in IoT smart city projects are optimal resource allocation, task scheduling, network, and resource slicing. The papers ([Dutta and Biswas, 2022](#); [Muthanna et al., 2022](#); [Nguyen et al., 2022](#); [Krishnan and Lim, 2021](#); [Nauman et al., 2021](#); [Cong et al., 2021](#); [Ali et al., 2020](#); [Muteba et al., 2020](#); [Gazori et al., 2020](#); [Chowdhury et al., 2019](#); [Gai and Qiu, 2018](#); [Zhu et al., 2021b](#); [He et al., 2020](#); [Hao et al., 2021](#)) solve this problem by using RL. The “SecOFF-FCIoT” scheme, presented by the authors uses Q-learning to determine whether the public or private cloud server should receive the workload ([Alli and Alam, 2019](#)).

Self-driving cars and unmanned automated vehicles (aerial and underwater) are also considered an important part of smart cities. Improvements to its implementation employ RL in multiple ways. Yang et al. dealt with the challenge of computation offloading and task scheduling in UAVs ([Yang et al., 2020](#)). The DRL algorithm, with

the time complexity of $O(n)$, improves the task scheduling efficiency compared to First Come First Server and Round Robin.

Papers [Rasheed et al. \(2020\)](#), [Hao et al. \(2021\)](#) and [Alqahtani and Hu \(2022\)](#) also use RL algorithms for various applications of unmanned automated vehicles. [Alqahtani and Hu \(2022\)](#) works on the issue of energy conservation by routing electric vehicles to supply energy to different users. [Rasheed et al. \(2020\)](#) improves the robustness of autonomous vehicle systems by using game theory. Two new DRL algorithms are used — each adversarial to the other. The authors [Chen et al. \(2021\)](#) optimized the privacy of smart vehicles by striking a balance between geolocation obfuscation and semantic security. A differential privacy scheme is used with RL to achieve this. This brings us to the topic of Federated Learning (FL) and differential privacy.

FL is an ML technique where decentralized algorithm training and testing is carried out on the private data of edge devices or servers known as participants, at the device itself, without ever needing to share the data with other participants or servers. The research article [Hao et al. \(2021\)](#) implements zero trust architecture and FL using A3C for providing ultra-reliable low latency communication services with resource slicing and scheduling on a 6G vehicular network. Articles [Chen et al. \(2021\)](#), [Wang et al. \(2020\)](#), [Ren et al. \(2022\)](#), [Ahmadi et al. \(2021\)](#), [Xu et al. \(2021\)](#), [Miao et al. \(2021\)](#) and [Zhan et al. \(2020\)](#) all combine FL and RL in different IoT environments.

Content-centric IoTs that stream multimedia is another hot topic where RL is being applied. IoT devices are fundamentally limited in energy and bandwidth ([Huang et al., 2018](#)). Content browsing and

Table 3

Datasets.

Dataset	Features	Advantages	Limitations
NSL-KDD (Caminero et al., 2019; Sethi et al., 2021; Suwannalai and Polprasert, 2020; Mohanty et al., 2021; Lopez-Martin et al., 2021; Ma and Shi, 2021; Hsu and Matsuoka, 2020; Dong et al., 2021; Shi and He, 2021; Xia et al., 2021; Muhati and Rawat, 2021; Lopez-Martin et al., 2020; Tavallaei et al., 2009)	125,973 training samples 22,544 test samples 41 features	A large number of samples.	Inconsistency in the number and type of attacks in the test set and training set. Some features were strings or characters.
AWID (Caminero et al., 2019; Lopez-Martin et al., 2021; Dong et al., 2021; Xia et al., 2021; Muhati and Rawat, 2021; Lopez-Martin et al., 2020; Kolias et al., 2016)	1,795,474 training samples 675,642 test samples 48 features	Largest and most comprehensive dataset collected in a real Wi-Fi network environment. No unknown attacks.	The value range of different features is quite different. Some features have null values, constant values, and network addresses.
UNSW-NB15 (Lopez-Martin et al., 2021; Hsu and Matsuoka, 2020; Muhati and Rawat, 2021)	175,341 training samples 82,332 testing samples 49 features	Large number of samples.	Class imbalance and class overlap.
ISOT-CID (Mohanty et al., 2021)	2.5 terabytes of data Web traffic generated by: 160 legitimate visitors (60 human + 100 robots)	This dataset consists of data obtained from various sources. Covers a wide variety of attack scenarios.	Essentially, raw data that have not been transformed, manipulated, or altered.
CICIDS2017 (Sethi et al., 2021; Lopez-Martin et al., 2021)	2.8 million traffic traces 15 classes 80 features	Variations of this dataset with the names of CICIDS and CICIDS2019 Lopez-Martin et al. (2021) are available.	Some of the TCP flows do not accurately reflect their underlying TCP connection. The lack of documentation concerning the flow construction and parameters of attacks.
DReLAB (Venturi et al., 2021)	Unique dataset for adversarial attacks on RL-based botnet Intrusion Detection Systems	Contains samples capable of evading the detection with less than 80 actions.	A clear definition of states, actions and flow is must, generate adversarial samples for the malicious flow.

streaming are made efficient by having an adaptive physical layer with adaptive rate and power control (Jagannath et al., 2019; Liu et al., 2018). Resource allocation, scheduling (Chowdhury et al., 2019), and network slicing are other methods that are used to make it more serviceable (He et al., 2020). (See Table 5.)

5. Research directions

Most of the research articles covered in the above Sections were developed for a specific use case. The authors of Tao et al. (2021) and Bouhamed et al. (2021) both have designed an RL-based IDS algorithm for UAVs. However, the state-action space in both cases varies. One model uses a DDPG double DNN approach and the other model uses DQN for attack classification. The generalization of the algorithms within the UAV application itself is challenging. The authors of Bouhamed et al. (2021) consider periods of offline learning for the UAVs. This can prove to be critical in an application where on-the-go learning is essential. Thus, the generalization of RL algorithms is a common challenge we have seen in many studies.

Fig. 9 summarizes the common topics for existing literature under the IDS, IPS, IoT, and IAM domains. Many topics like computation and data offloading (Huang et al., 2020), anti-jamming (Aref et al., 2017), interference avoidance (Suwannalai and Polprasert, 2020), navigation (Rasheed et al., 2020), and authentication (Xiao et al., 2021)

also fall under the category of networks. An important issue that needs to be addressed in these applications is the high generalization ability for the high-variance and dynamic network traffic conditions commonly encountered in all networks. Often, the performance of ML algorithms fluctuates when the network conditions change. For example, an algorithm that considers signal strength as the threshold for authenticating a user could fail in poor network conditions.

The robustness of the algorithm is a key challenge in several applications. Sensitive tasks and applications like autonomous navigation (Rasheed et al., 2020) and smart home security (Heartfield et al., 2021) need schemes, which are resilient to outside factors and parameters. Lapses in the reliability and robustness in such cases can result in catastrophic events.

As much as robustness is important, preserving privacy is another important facet of the cybersecurity shield that we discuss here. Few papers address the issue of privacy preservation using Homomorphic Encryption (HE) or Fully Homomorphic Encryption (FHE). HE is a type of encryption that allows users to directly compute encrypted data. There is no need to decrypt before performing computations and operations, thus preserving the privacy of the data (Crawford et al., 2018). The keyword search result for 'homomorphic encryption reinforcement learning' on IEEEExplore accessed on 14th May 2022, results only six results. There is tremendous scope for novel research in this direction.

Table 4
Summary of reinforcement learning methods used in IAM.

Reference	Model/ Application	Learning algorithm	Agent	States	Actions	Reward	Future Scope
Ferdowsi and Saad (2019)	DW-LSTM for authentication of IoT Signals	Dynamic Watermarking deep LSTM (RNN), DQN	Gateway	N dimensional attacker action sequence	0 — IoT not under attack; 1 — under attack; 2 — the gateway did not authenticate this IoT.	Utility based on payoff	The watermarked signal from IoTs is written as a function. It has two main components — relative power of the pseudo-noise signal to the original signal and number of samples (frame length) of IoTs original signal used to hide a single bit. If we select these two to be large values, bit extraction error becomes null. However, it leads to IoT gateways having to incur latency and computational costs.
Xiao et al. (2015, 2016)	Spoofing detection	Q-learning Xiao et al. (2015), Dyna-Q Xiao et al. (2016)	Receiver in a single hop radio network	False alarm rate and miss detection rate of authentication	Set authentication threshold from among K levels	Sum utility	(1). Increase in number of nodes increases the miss detection rate. 2). Methods to ensure that average utility of the receiver does not decrease with the number of nodes
Gao et al. (2020)	Intelligent Spoofing Attack	Finite MDP, Q-learning	Attacker	Observation and hidden states	Launch attack or not	Channel coefficient	(1). Investigate the cooperative PHY-layer authentication scheme in mobile WSN. (2). Review the effects of non-Gaussian noise on measurements in the practical systems.
Xu et al. (2019)	CAN PHY-Authentication Scheme	Q-learning	ECU	State set — previous estimated false alarm rate, miss detection rate, attack rate and current message priority.	Protection level and test threshold	Sum utility	Proposed method provides medium level protection to authenticity of messages. If advanced protection is required, it comes increased computation overhead and authentication latency. Review methods to analyze voltage tones to provide advanced protection.
Lu et al. (2020)	DRLPA	Dyna-Q, Transfer learning, CNN	Mobile device in VANET	Quantized false alarm rates, miss detection rates, quantized estimated attack rates, quantized priority levels of the current packet under test.	Choose authentication policy — authentication modes and authentication parameters	Utility given by packet priority, the authentication cost and authentication accuracy	Explore DL combinations with the proposed scheme improve authentication performance and optimization time.
Xiao et al. (2019)	RAP, DRAP	Q-learning with hotbooting technique, CNN, PDP	Underwater sensor	Previous spoofing detection performance and the predicted attack strength	Choose authentication threshold	Long term utility	Using DNN reduces the false alarm rate by 38.9% and increases the utility by 36.7%. Experiment with more different types of neural networks to see if this can be improved further.
Ren et al. (2022)	PICRL	Q-learning and Differential Privacy	Crowdsourcing IoT application	The participants selected by the system platform are the state space.	Act of random participant selection	R = Negative value; if payment > budget; else R = increase of the system utility.	(1). Increase participant privacy exposure by adding an incentive mechanism or game theory. (2). Calculate payment of participants from other factors like privacy exposure degree, trust value and data amount. (3). Evaluate online trust mechanism to reduce response times using AI. (4). Explore trust value evaluation for the dynamic arrival and joining of new participants.

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Table 4 (continued).

Reference	Model/ Application	Learning algorithm	Agent	States	Actions	Reward	Future Scope
Ahmadi et al. (2021)	DQRE-SCnet	Q-learning, Spectral Clustering, Multi-layer neural network with ReLu and SoftMax activation functions	Data privacy in FL and training-IoT	Compare a single x data with several classes.	Place data in an appropriate class.	A function of mean squares error of instances from the training data.	Quantification of potential consequences like bias while developing the algorithm or process and employing methods to minimize them.

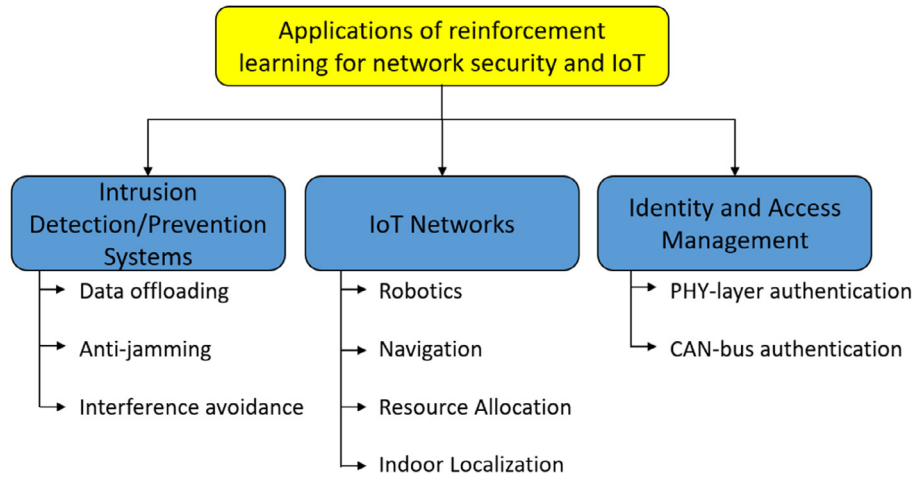


Fig. 9. A taxonomy of IDS, IoT and IAM systems using reinforcement learning.

Another study reviewed in this survey uses HE to achieve data privacy. A novel approach for preserving data privacy was proposed by the authors of Miao et al. (2021). The sensitive task decomposition is done by a hierarchical asynchronous FL framework. RL combined with RL is a new research area. FL is an ML method that enables collaborative learning from mobiles, servers, and data centers without sharing the data itself.

There is a potential way ahead if we consider research in FL RL. DDQN is used in Xue et al. (2021) to identify a stable and sequential clinical treatment policy to assist clinicians in making proper treatment decisions. From there, a fully decentralized federated framework collects the knowledge acquired by the models of DDQN while ensuring the privacy of the personal health information. Researchers must pay special attention to the robustness of the algorithm since the results are sensitive to the reward function based on sequential organ failure assessment scores.

There is another combination, that of Blockchain and RL, which has many open challenges and application areas. We reviewed a survey study that investigated the blockchain and DRL applications in IIoT (Wu et al., 2021). However, the true use of the capabilities of blockchain lies in using them for authentication (Zhu et al., 2021a). Its full potential can be explored by combining it with RL algorithms for IAM applications (Tao et al., 2021; Alli and Alam, 2019; Hao et al., 2021). We discuss it in more detail below.

A common issue seen throughout the literature survey, barring a handful of papers, is the lack of reproducibility of RL-based algorithms. A better description of the environment, agent configuration, action, and state space is needed to improve reproducibility. This will, in turn, help with accountability and interpretability. Practical implementations of the methods must be made accessible to the wider research community to encourage a better understanding of the 'why's' and 'how's' of the models.

The future scope of individual research articles can be highly varied. However, we do observe a common thread looking at the high-level survey of each domain.

The shortcomings and limitations of the current literature are captured in Fig. 10 above. It is summarized under the following sub-topics:

5.1. Insufficiency of standard reinforcement learning

RL by itself can prove to be insufficient mainly because state space can become massive very quickly. A research article (Lopez-Martin et al., 2021) recommends the use of a DNN to make the action space more manageable. In Tao et al. (2021), the future research directions recommend using a blockchain in integration with the presented DDPG scheme secure IDS classifiers in UAVs. Alli and Alam (2019) also recommends a combination of blockchain to ensure secure computation offloading. The authors Hao et al. (2021) recommend the usage of blockchain for strengthening the Zero Trust Architecture of 6G networks. Lastly, Ren et al. (2022) recommends making privacy exposure more attractive for participants by adding an incentive mechanism or game theory.

5.2. Adversarial reinforcement learning algorithms

An upcoming trend is where attackers operate on AI and ML technologies and collect data over a period to make covert attacks. We have seen above that there are algorithms and game theories developed where the cybercriminals learn about the vulnerable devices and nodes using RL. It is not just attackers, which use RL, there are attacks where the target is the RL policy itself (Huang et al., 2019; Hadfield-Menell et al., 2017; Aref et al., 2017; Anderson et al., 2017).

5.3. Prioritized experience replay and actor-critic algorithms

Experience replay allows the RL agents to reuse and remember past experiences. All the experiences are unequally significant. Several rare events are most significant, and not necessarily anomalies. Replaying them is beneficial. It often happens that some of the old and rare

Table 5

Summary of reinforcement learning methods used in IoT.

Reference	Model/ Application	Learning algorithm	Agent	States	Actions	Reward	Future Scope
Krishnan and Lim (2021)	RL-based mobile sink model	Q-Learning	Mobile Sink	Cluster heads (CH) location	Move from current CH location to next CH location	The lesser the travel distance to the next CH, higher the reward	(1). Improving the resiliency of a communication channel in the presence of mobility, obstacles, and noise. (2). Integration with other RL algorithms and multiple devices.
Rasheed et al. (2020)	NDRL LSTM-GAN	Game Theory, Q-learning	Vehicles	Anticipated distance variation	Move the vehicle by some distance	Maximize Q-value	(1). Introduce the lateral dynamics that will make the system model non-linear. (2). Improve 'car following model' by building it on driving strategies. (3). To improve the estimation error analysis of stochastic systems. (4). Develop new attack detection mechanisms.
Alli and Alam (2019)	SecOFF-FCIoT	Q-learning,	Mobile IoT devices	All available VMs	Choose Virtual Machine (VM) from the existing VMs	Rewards obtained denotes the cloud servers current state (running, waiting, busy etc.)	(1). Simulate a large-scale environment. (2). Construct a cluster-based network (3). Attempt to reduce energy consumption in IoTs. (4). Integrate with blockchain.
Liu et al. (2018)	HIF-DRL	DQN	User of High Frequency channel communication	High frequencies	Frequency selection	The function of cumulative jamming or interference experienced by each channel.	Design multi-user HIF-DRL algorithm
Yang et al. (2020)	DRL-aided task scheduling	DQN	UAV, Task scheduling	Computing resources	Scheduled task was moved to the appropriate position	Minimize the average slowdown in the UAVs.	The model assumes that all IoT nodes are fixed on the ground and all UAVs fly at a fixed altitude. Future research into the performance when the system is truly dynamic is encouraged.
Miao et al. (2021)	DRNS (Deep Reinforcement Node Selection)	DDPG, Q-learning	IoT data sharing	System state $s(t) = [Li(t), Ci(t), \lambda i(t)]$ Li - position of each node; Ci - connection state of node; $\lambda i(t)$ - node selection state.	Node selection	The function of the local test accuracy of each node and the number of data nodes.	Homomorphic encryption algorithm — Paillier encryption algorithm was used. The improvement of speed of computation should be investigated.
Chen et al. (2021)	Optimized Differential Privacy Scheme with RL	IoT vehicle trajectory	Vehicle	The state at the time stamp t , including geographic location dt , semantic location ct .	Increasing, decreasing, or maintaining the privacy budget.	The function of seg-similarity and semantic security. Goal — when the protected trajectory and the original trajectory have a relatively large similarity, the semantic loss should be relatively small.	Research on the possibility of privacy gain is large. This could lead to the semantic security risk being high.

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experiences are ignored. RL agents must be programmed with the intelligence to employ an efficient sampling strategy for the events. Rare, old, yet important experiences are replayed for learning. There is tremendous scope and little work done in this domain (Caminero et al., 2019).

5.4. Access management and authentication

RL-based authentication at the application layer also has not been explored much. PHY-layer authentication is the most popular; however, all the PHY-layer-based authentication algorithms recommend that

Table 5 (continued).

Reference	Model/ Application	Learning algorithm	Agent	States	Actions	Reward	Future Scope
Hao et al. (2021)	URLLC	A3C	Computation offloading in IoT vehicles	The state is a function of vehicular tasks, which need to be offloaded, remaining travel time, computation resource states of an edge server, a convergence server, and a cloud server and bandwidth.	Server selection when vehicular tasks are offloaded	The reward is such that it minimizes the cost function of the actions.	(1). Use blockchain to improve the zero trust architecture for 6G vehicular services. (2). Explore the usage of the zero-knowledge proof method to improve authentication security.

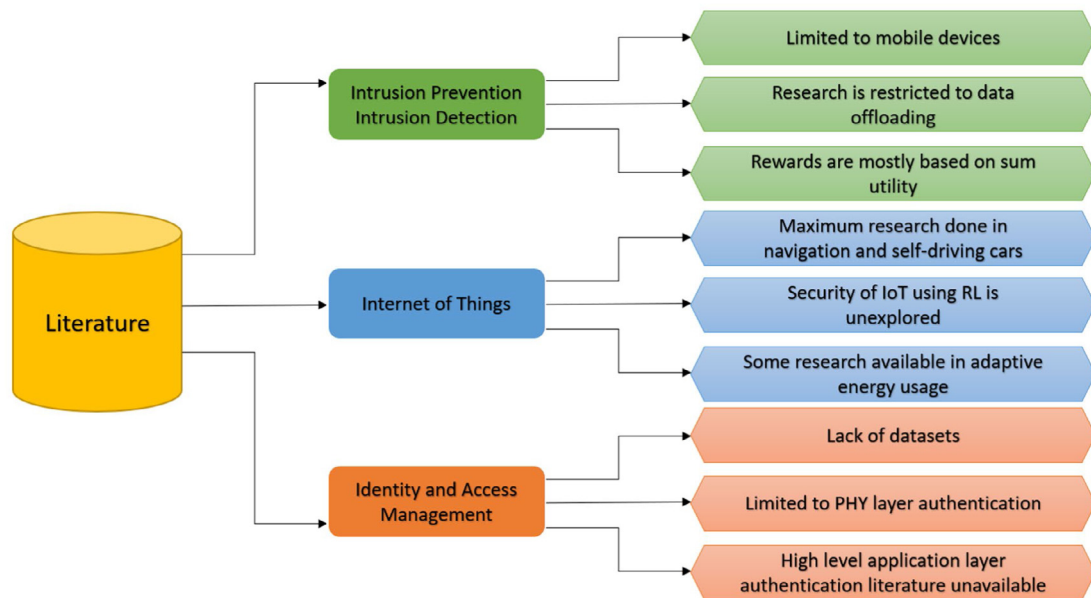


Fig. 10. Shortcomings of existing literature.

higher-level authentication must be implemented. The future scope of Hao et al. (2021) proposes to explore zero trust architecture to ensure authentication is secure. Also, Xiao et al. (2015, 2016) encourage researchers to investigate the cooperative PHY-layer authentication schemes in mobile WSNs and other such networks. Access assignment to resources has barely been touched upon. We plan on exploring this area further and evaluating its application and implementation.

In the last five years, RL is being explored while implementing newer technologies like IoT. Many researchers are implementing RL in established technologies such as cybersecurity. Authentication and RL is a contemporary area where many publications are not available. During the literature survey for this paper, the authors identified very few patents filed for RL and cybersecurity. Most of the patents were from China. The authors plan to include this information in future research.

6. Conclusion

In a nutshell, RL-based intelligent systems are economic, robust, pervasive, and dynamic. To conclude, it is worth revisiting the overarching goal of this research: identify research areas and state-of-the-art methods using RL for various aspects of cybersecurity. We have covered

numerous models and proposed algorithms within the cybersecurity domain. We introduced RL algorithms used in the cybersecurity domain by categorizing them into three applications. This extensive survey paper presents the IDS, IPS, IoT, and IAM applications where reinforcement learning is used. The latest applications of RL in descending order are IDS classifiers, anomaly detectors (Sethi et al., 2021, 2020; Tao et al., 2021; Gu et al., 2020; Puzanov et al., 2020; Bouhamed et al., 2021; Suwannalai and Polprasert, 2020; Heartfield et al., 2021; Ma and Shi, 2021; Otoum et al., 2021; Mohanty et al., 2021; Lopez-Martin et al., 2021), resource allocation, task scheduling (Dutta and Biswas, 2022; Muthanna et al., 2022; Nguyen et al., 2022; Krishnan and Lim, 2021; Nauman et al., 2021; Cong et al., 2021; Ali et al., 2020; Muteba et al., 2020; Gazori et al., 2020; Chowdhury et al., 2019; Gai and Qiu, 2018; Zhu et al., 2021b; He et al., 2020), computation offloading (Huang et al., 2020; Yang et al., 2020), FL, differential privacy (Chen et al., 2021; Wang et al., 2020; Ren et al., 2022; Ahmadi et al., 2021; Xu et al., 2021; Miao et al., 2021; Zhan et al., 2020), smart city (Goyal et al., 2021; Wang et al., 2021; Ren et al., 2021; Bu and Wang, 2019), PHY-layer authentication (Xiao et al., 2015, 2016; Tao et al., 2021; Xiao et al., 2017; Gao et al., 2020; Xiao et al., 2018b; Lu et al., 2020; Greenberg et al., 2020; Xiao et al., 2021). This paper examined several databases used to train RL-based IDS and IPS systems.

We proposed a set of evaluation criteria and used them to compare and analyze existing methods in various application categories. Following a thorough review, we proposed some open questions and made recommendations for future research.

The issue of large unwieldy state spaces is solved by combining RL with other techniques like blockchain, DNN, or FL. This allows efficiency to improve. Such as, energy savings of 22.75% are achieved when employing RL for autonomous vehicle routing compared with the particle swarm optimization technique (Alqahtani and Hu, 2022). Using DRL to reduce computation time and time to convergence should also be explored in future research papers.

We even reviewed other surveys which recommend solving problems in large-scale deployments using RL. Exploring applications of RL in zero trust architecture and secure authentication is also recommended (Gaba et al., 2022). Authentication, access management, blockchain, FL, privacy preservation, and FHE are some topics with tremendous potential for exploration and research in the domain of reinforcement learning. The Section on future scope in this paper provides ample guidance for researchers to continue their quest for making reinforcement learning-based models more secure, fail-safe, stable, intuitive, and overall human-like.

CRedit authorship contribution statement

Amrin Maria Khan Adawadkar: Conceptualization, Methodology, Resources, Writing – original draft, Investigation. **Nilima Kulkarni:** Supervision, Data curation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Agarwal, R., Schwarzer, M., Castro, P.S., Courville, A., Bellemare, M.G., 2021. Deep reinforcement learning at the edge of the statistical precipice. In: *Advances in Neural Information Processing Systems*. <http://dx.doi.org/10.48550/arXiv.2108.13264>.
- Ahmadi, M., Taghaviashidzadeh, A., Javaheri, D., Masoumian, A., Jafarzadeh Ghouschi, S., Pourasad, Y., 2021. DQRE-SCnet: A novel hybrid approach for selecting users in federated learning with deep-Q-reinforcement learning based on spectral clustering. *J. King Saud Univ., Comput. Inf. Sci.* <http://dx.doi.org/10.1016/j.jksuci.2021.08.019>.
- Ali, R., Kim, B., Kim, S.W., Kim, H.S., Ishmanov, F., 2020. (ReLBT): A Reinforcement learning-enabled listen before talk mechanism for LTE-LAA and Wi-Fi coexistence in IoT. *Comput. Commun.* <http://dx.doi.org/10.1016/j.comcom.2019.11.055>.
- Alli, A.A., Alam, M.M., 2019. SecOFF-FCIoT: ML based secure offloading in fog-cloud of things for smart city applications. *Internet of Things* <http://dx.doi.org/10.1016/j.iot.2019.100070>.
- Alqahtani, M., Hu, M., 2022. Dynamic energy scheduling and routing of multiple electric vehicles using deep reinforcement learning. *Energy* <http://dx.doi.org/10.1016/j.energy.2021.122626>.
- Anderson, H.S., Filar, B., Roth, P., 2017. Evading ML Malware Detection. *BlackHat, DC*.
- Anon, General data protection regulation (EU) cor. OJ L 127, 23, 2018.
- Aref, M.A., Jayaweera, S.K., Machuzak, S., 2017. Multi-agent reinforcement learning based cognitive anti-jamming. In: *IEEE Wireless Communications and Networking Conference, WCNC*. <http://dx.doi.org/10.1109/WCNC.2017.7925694>.
- Arulkumaran, K., Deisenroth, M.P., Brundage, M., Bharath, A.A., 2017. Deep reinforcement learning: A brief survey. *IEEE Signal Process. Mag.* <http://dx.doi.org/10.1109/MSP.2017.2743240>.
- Bellman, R., 1952. On the theory of dynamic programming. *Proc. Natl. Acad. Sci.* <http://dx.doi.org/10.1073/pnas.38.8.716>.
- Bennaceur, K., Sahraoui, Z., Labed, A., Ahmed-Nacer, M., 2019. Training function stability in anomaly intrusion detection based deep learning. In: *2019 International Conference on Control, Automation and Diagnosis, ICCAD 2019 - Proceedings*. <http://dx.doi.org/10.1109/ICCAD46983.2019.9037956>.
- Bouhamed, O., Bouachir, O., Alokaily, M., Ridhawi, I. Al, 2021. Lightweight IDS for UAV networks: A periodic deep reinforcement learning-based approach. In: *Proceedings of the IM 2021-2021 IFIP/IEEE International Symposium on Integrated Network Management*.
- Bu, F., Wang, X., 2019. A smart agriculture IoT system based on deep reinforcement learning. *Future Gener. Comput. Syst.* <http://dx.doi.org/10.1016/j.future.2019.04.041>.
- Buduru, A.B., Yau, S.S., 2015. An effective approach to continuous user authentication for touch screen smart devices. In: *Proceedings - 2015 IEEE International Conference on Software Quality, Reliability and Security, QRS 2015*. <http://dx.doi.org/10.1109/QRS.2015.40>.
- Camirero, G., Lopez-Martin, M., Carro, B., 2019. Adversarial environment reinforcement learning algorithm for intrusion detection. *Comput. Netw.* <http://dx.doi.org/10.1016/j.comnet.2019.05.013>.
- Chen, X., Zhang, T., Shen, S., Zhu, T., Xiong, P., 2021. An optimized differential privacy scheme with reinforcement learning in VANET. *Comput. Secur.* <http://dx.doi.org/10.1016/j.cose.2021.102446>.
- Chowdhury, A., Raut, S.A., Narman, H.S., 2019. DA-DRLS: Drift adaptive deep reinforcement learning based scheduling for IoT resource management. *J. Netw. Comput. Appl.* <http://dx.doi.org/10.1016/j.jnca.2019.04.010>.
- Cong, P., Zhang, Y., Liu, Z., Baker, T., Tawfik, H., Wang, W., Xu, K., Li, R., Li, F., 2021. A deep reinforcement learning-based multi-optimality routing scheme for dynamic IoT networks. *Comput. Netw.* <http://dx.doi.org/10.1016/j.comnet.2021.108057>.
- Crawford, J.L.H., Gentry, C., Halevi, S., Platt, D., Shoup, V., 2018. Doing real work with FHE: The case of logistic regression. In: *Proceedings of the ACM Conference on Computer and Communications Security*. <http://dx.doi.org/10.1145/3267973.3267974>.
- Dong, S., Xia, Y., Peng, T., 2021. Network abnormal traffic detection model based on semi-supervised deep reinforcement learning. *IEEE Trans. Netw. Serv. Manag.* <http://dx.doi.org/10.1109/TNSM.2021.3120804>.
- Dutta, H., Biswas, S., 2022. Distributed reinforcement learning for scalable wireless medium access in IoTs and sensor networks. *Comput. Netw.* <http://dx.doi.org/10.1016/j.comnet.2021.108662>.
- Elderman, R., Pater, L.J.J., Thie, A.S., Drugan, M.M., Wiering, M.A., 2017. Adversarial reinforcement learning in a cyber security simulation. In: *ICAART 2017 - Proceedings of the 9th International Conference on Agents and Artificial Intelligence*. <http://dx.doi.org/10.5220/0006197105590566>.
- Erdódi, L., Somervoll, Å.Å., Zennaro, F.M., 2021. Simulating SQL injection vulnerability exploitation using Q-learning reinforcement learning agents. *J. Inform. Secur. Appl.* <http://dx.doi.org/10.1016/j.jisa.2021.102903>.
- Fang, H., Wang, X., Tomasin, S., 2019. ML for intelligent authentication in 5G and beyond wireless networks. *IEEE Wirel. Commun.* <http://dx.doi.org/10.1109/MWC.001.1900054>.
- Ferdowsi, A., Saad, W., 2019. Deep learning for signal authentication and security in massive Internet-of-Things systems. *IEEE Trans. Commun.* <http://dx.doi.org/10.1109/TCOMM.2018.2878025>.
- Fernandez, G.C., Xu, S., 2019. A case study on using deep learning for network intrusion detection. In: *Proceedings - IEEE Military Communications Conference MILCOM*. <http://dx.doi.org/10.1109/MILCOM47813.2019.9020824>.
- Gaba, G.S., Hedabou, M., Kumar, P., Braeken, A., Liyanage, M., Alazab, M., 2022. Zero knowledge proofs based authenticated key agreement protocol for sustainable healthcare. *Sustain. Cities Soc.* <http://dx.doi.org/10.1016/j.scs.2022.103766>.
- Gai, K., Qiu, M., 2018. Optimal resource allocation using reinforcement learning for IoT content-centric services. *Appl. Soft Comput. J.* <http://dx.doi.org/10.1016/j.asoc.2018.03.056>.
- Gao, Y., Liu, Y., Jin, Y., Chen, J., Wu, H., 2018. A novel semi-supervised learning approach for network intrusion detection on cloud-based robotic system. *IEEE Access*. <http://dx.doi.org/10.1109/ACCESS.2018.2868171>.
- Gao, N., Ni, Q., Feng, D., Jing, X., Cao, Y., 2020. Physical layer authentication under intelligent spoofing in wireless sensor networks. *Signal Process.* <http://dx.doi.org/10.1016/j.sigpro.2019.107272>.
- Gazori, P., Rahbari, D., Nickray, M., 2020. Saving time and cost on the scheduling of fog-based IoT applications using deep reinforcement learning approach. *Future Gener. Comput. Syst.* <http://dx.doi.org/10.1016/j.future.2019.09.060>.
- Goyal, H.R., Ghanshala, K.K., Sharma, S., 2021. Flash flood risk management modeling in indian cities using IoT based reinforcement learning. In: *Materials Today: Proceedings*. <http://dx.doi.org/10.1016/j.matpr.2021.01.072>.
- Greenberg, A., Burgess, M., Grey, W., Hall, J., Giordano, S., Anastasio, M., Nield, D., 2020. Hackers remotely kill a jeep on the highway - with me in it. [online] WIRED. Available at: <https://www.wired.com/2015/07/hackers-remotely-kill-jeep-highway/>. (Accessed 8 2020).
- Gu, T., Abhishek, A., Fu, H., Zhang, H., Basu, D., Mohapatra, P., 2020. Towards learning-automation IoT attack detection through reinforcement learning. In: *Proceedings - 21st IEEE International Symposium on a World of Wireless, Mobile and Multimedia Networks, WoWMoM 2020*. <http://dx.doi.org/10.1109/WoWMoM49955.2020.00029>.
- Hadfield-Menell, D., Dragan, A., Abbeel, P., Russell, S., 2017. The off-switch game. In: *IJCAI International Joint Conference on Artificial Intelligence*. <http://dx.doi.org/10.24963/ijcai.2017/32>.
- Hao, M., Ye, D., Wang, S., Tan, B., Yu, R., 2021. URLLC resource slicing and scheduling for trustworthy 6G vehicular services: A federated reinforcement learning approach. *Phys. Commun.* <http://dx.doi.org/10.1016/j.phycom.2021.101470>.
- He, X., Wang, K., Huang, H., Miyazaki, T., Wang, Y., Guo, S., 2020. Green resource allocation based on deep reinforcement learning in content-centric IoT. *IEEE Trans. Emerg. Top. Comput.* <http://dx.doi.org/10.1109/TETC.2018.2805718>.

- Heartfield, R., Loukas, G., Bezemskij, A., Panaousis, E., 2021. Self-configurable cyber-physical intrusion detection for smart homes using reinforcement learning. *IEEE Trans. Inf. Forensics Secur.* <http://dx.doi.org/10.1109/TIFS.2020.3042049>.
- Hsu, Y.F., Matsuoka, M., 2020. A deep reinforcement learning approach for anomaly network intrusion detection system. In: *Proceedings - 2020 IEEE 9th International Conference on Cloud Networking, CloudNet 2020*. <http://dx.doi.org/10.1109/CloudNet51028.2020.9335796>.
- Hu, J., Wellman, M.P., 2004. Nash Q-learning for general-sum stochastic games. *J. Mach. Learn. Res.* <http://dx.doi.org/10.1162/1532443041827880>.
- Huang, L., Bi, S., Zhang, Y.J.A., 2020. Deep reinforcement learning for online computation offloading in wireless powered mobile-edge computing networks. *IEEE Trans. Mob. Comput.* <http://dx.doi.org/10.1109/TMC.2019.2928811>.
- Huang, S., Papernot, N., Goodfellow, I., Duan, Y., Abbeel, P., 2019. Adversarial attacks on neural network policies. In: *5th International Conference on Learning Representations, ICLR 2017 - Workshop Track Proceedings*.
- Huang, T., Zhang, R.X., Zhou, C., Sun, L., 2018. QARC: Video quality aware rate control for real-time video streaming based on deep reinforcement learning. In: *MM 2018 - Proceedings of the 2018 ACM Multimedia Conference*. <http://dx.doi.org/10.1145/3240508.3240545>.
- Iacovazzi, A., Baiocchi, A., 2014. Internet traffic privacy enhancement with masking: Optimization and tradeoffs. *IEEE Trans. Parallel Distrib. Syst.* <http://dx.doi.org/10.1109/TPDS.2013.42>.
- Jagannath, J., Polosky, N., Jagannath, A., Restuccia, F., Melodia, T., 2019. ML for wireless communications in the Internet of Things: A comprehensive survey. *Ad Hoc Netw.* <http://dx.doi.org/10.1016/j.adhoc.2019.101913>.
- Khan, S., Shiraz, M., Boroumand, L., Gani, A., Khan, M.K., 2017. Towards port-knocking authentication methods for mobile cloud computing. *J. Netw. Comput. Appl.* <http://dx.doi.org/10.1016/j.jnca.2017.08.018>.
- Knirsch, F., Eibl, G., Engel, D., 2018. Error-resilient masking approaches for privacy preserving data aggregation. *IEEE Trans. Smart Grid.* <http://dx.doi.org/10.1109/TSG.2016.2630803>.
- Kolias, C., Kambourakis, G., Stavrou, A., Gritzalis, S., 2016. Intrusion detection in 802.11 networks: Empirical evaluation of threats and a public dataset. *IEEE Commun. Surv. Tutor.* <http://dx.doi.org/10.1109/COMST.2015.2402161>.
- Kotani, G., Sekiya, Y., 2019. Unsupervised scanning behaviours detection based on distribution of network traffic features using robust autoencoders. In: *IEEE International Conference on Data Mining Workshops, ICDMW*. <http://dx.doi.org/10.1109/ICDMW.2018.00013>.
- Krishnan, M., Lim, Y., 2021. Reinforcement learning-based dynamic routing using mobile sink for data collection in WSNs and IoT applications. *J. Netw. Comput. Appl.* <http://dx.doi.org/10.1016/j.jnca.2021.103223>.
- Li, Y., Quevedo, D.E., Dey, S., Shi, L., 2017. SINR-based DoS attack on remote state estimation: A game-theoretic approach. *IEEE Trans. Control Netw. Syst.* <http://dx.doi.org/10.1109/TCNS.2016.2549640>.
- Ling, M.H., Yau, K.L.A., Qadir, J., Poh, G. Sen, Ni, Q., 2015. Application of reinforcement learning for security enhancement in cognitive radio networks. *Appl. Soft Comput. J.* <http://dx.doi.org/10.1016/j.asoc.2015.09.017>.
- Liu, X., Xu, Y., Cheng, Y., Li, Y., Zhao, L., Zhang, X., 2018. A heterogeneous information fusion deep reinforcement learning for intelligent frequency selection of HF communication. *China Commun.* <http://dx.doi.org/10.1109/CC.2018.8456453>.
- Lopez-Martin, M., Carro, B., Sanchez-Esguevillas, A., 2020. Application of deep reinforcement learning to intrusion detection for supervised problems. *Expert Syst. Appl.* <http://dx.doi.org/10.1016/j.eswa.2019.112963>.
- Lopez-Martin, M., Sanchez-Esguevillas, A., Arribas, J.I., Carro, B., 2021. Network intrusion detection based on extended RBF neural network with offline reinforcement learning. *IEEE Access.* <http://dx.doi.org/10.1109/ACCESS.2021.3127689>.
- Lu, X., Xiao, L., Xu, T., Zhao, Y., Tang, Y., Zhuang, W., 2020. Reinforcement learning based PHY authentication for VANETs. *IEEE Trans. Veh. Technol.* <http://dx.doi.org/10.1109/TVT.2020.2967026>.
- Ma, X., Shi, W., 2021. AESMOTE: Adversarial reinforcement learning with SMOTE for anomaly detection. *IEEE Trans. Netw. Sci. Eng.* <http://dx.doi.org/10.1109/TNSE.2020.3004312>.
- Maeda, R., Mimura, M., 2021. Automating post-exploitation with deep reinforcement learning. *Comput. Secur.* <http://dx.doi.org/10.1016/j.cose.2020.102108>.
- Marir, N., Wang, H., Feng, G., 2019. Unsupervised feature learning with distributed stacked denoising sparse autoencoder for abnormal behavior detection using apache spark. In: *Proceedings of the 2nd IEEE International Conference on Knowledge Innovation and Invention 2019, ICKII 2019*. <http://dx.doi.org/10.1109/ICKII46306.2019.9042645>.
- Miao, Q., Lin, H., Wang, X., Hassan, M.M., 2021. Federated deep reinforcement learning based secure data sharing for Internet of Things. *Comput. Netw.* <http://dx.doi.org/10.1016/j.comnet.2021.108327>.
- Mirza, A.H., Cosan, S., 2018. Computer network intrusion detection using sequential LSTM neural networks autoencoders. In: *26th IEEE Signal Processing and Communications Applications Conference, SIU 2018*. <http://dx.doi.org/10.1109/SIU.2018.8404689>.
- Mnih, V., Badia, A.P., Mirza, L., Graves, A., Harley, T., Lillicrap, T.P., Silver, D., Kavukcuoglu, K., 2016. Asynchronous methods for deep reinforcement learning. In: *33rd International Conference on ML, ICML 2016*.
- Mohanty, D., Sethi, K., Prasath, S., Rout, R.R., Bera, P., 2021. Intelligent intrusion detection system for smart grid applications. In: *2021 International Conference on Cyber Situational Awareness, Data Analytics and Assessment, CyberSA 2021*. <http://dx.doi.org/10.1109/CyberSA52016.2021.9478200>.
- Muhati, E., Rawat, D.B., 2021. Asynchronous advantage actor-critic (A3C) learning for cognitive network security. In: *Proceedings of 2021 Third IEEE International Conference on Trust, Privacy and Security in Intelligent Systems and Applications, TPS-ISA*. <http://dx.doi.org/10.1109/TPSISA52974.2021.00012>.
- Muteba, K.F., Djouani, K., Olwal, T.O., 2020. Deep reinforcement learning based resource allocation for narrowband cognitive radio-IoT systems. In: *Procedia Computer Science*. <http://dx.doi.org/10.1016/j.procs.2020.07.046>.
- Muthanna, M.S.A., Muthanna, A., Rafiq, A., Hammoudeh, M., Alkanhel, R., Lynch, S., Abd El-Latif, A.A., 2022. Deep reinforcement learning based transmission policy enforcement and multi-hop routing in QoS aware LoRa IoT networks. *Comput. Commun.* <http://dx.doi.org/10.1016/j.comcom.2021.11.010>.
- Nauman, A., Jamshed, M.A., Ali, R., Cengiz, K., Zulfarnain, S.W., 2021. Reinforcement learning-enabled intelligent device-to-device (I-D2D) communication in narrow-band Internet of Things (NB-IoT). *Comput. Commun.* <http://dx.doi.org/10.1016/j.comcom.2021.05.007>.
- Nguyen, T.G., Phan, T.V., Hoang, D.T., Nguyen, H.H., Le, D.T., 2022. DeepPlace: Deep reinforcement learning for adaptive flow rule placement in software-defined IoT networks. *Comput. Commun.* <http://dx.doi.org/10.1016/j.comcom.2021.10.006>.
- Nguyen, T.T., Reddi, V.J., 2021. Deep reinforcement learning for cyber security. *IEEE Trans. Neural Netw. Learn. Syst.* <http://dx.doi.org/10.1109/TNNLS.2021.3121870>.
- Nugroho, E.P., Djatna, T., Sitanggang, I.S., Buono, A., Hermadi, I., 2020. A review of intrusion detection system in IoT with ML approach: Current and future research. In: *2020 6th International Conference on Science in Information Technology: Embracing Industry 4.0: Towards Innovation in Disaster Management, ICSITech 2020*. <http://dx.doi.org/10.1109/ICSITech49800.2020.9392075>.
- Otoun, S., Guizani, N., Mouftah, H., 2021. Federated reinforcement learning-supported IDS for IoT-steered healthcare systems. In: *IEEE International Conference on Communications*. <http://dx.doi.org/10.1109/ICC42927.2021.9500698>.
- Pamukov, M.E., Poulkov, V.K., 2017. Multiple negative selection algorithm: Improving detection error rates in IoT intrusion detection systems. In: *Proceedings of the 2017 IEEE 9th International Conference on Intelligent Data Acquisition and Advanced Computing Systems: Technology and Applications, IDAACS 2017*. <http://dx.doi.org/10.1109/IDAACS.2017.8095140>.
- Puzanov, A., Zhang, S., Cohen, K., 2020. Deep reinforcement one-shot learning for artificially intelligent classification in expert aided systems. *Eng. Appl. Artif. Intell.* <http://dx.doi.org/10.1016/j.engappai.2020.103589>.
- Rasheed, I., Hu, F., Zhang, L., 2020. Deep reinforcement learning approach for autonomous vehicle systems for maintaining security and safety using LSTM-GAN. *Veh. Commun.* <http://dx.doi.org/10.1016/j.vehcom.2020.100266>.
- Ren, Y., Liu, W., Liu, A., Wang, T., Li, A., 2022. A privacy-protected intelligent crowdsourcing application of IoT based on the reinforcement learning. *Future Gener. Comput. Syst.* <http://dx.doi.org/10.1016/j.future.2021.09.003>.
- Ren, T., Niu, J., Cui, J., Ouyang, Z., Liu, X., 2021. An application of multi-objective reinforcement learning for efficient model-free control of canals deployed with IoT networks. *J. Netw. Comput. Appl.* <http://dx.doi.org/10.1016/j.jnca.2021.103049>.
- Ring, M., Wunderlich, S., Scheuring, D., Landes, D., Hotho, A., 2019. A survey of network-based intrusion detection data sets. *Comput. Secur.* <http://dx.doi.org/10.1016/j.cose.2019.06.005>.
- Rustam, Z., Talita, A.S., 2018. Fuzzy kernel robust clustering for anomaly based intrusion detection. In: *Proceedings of the 3rd International Conference on Informatics and Computing, ICIC 2018*. <http://dx.doi.org/10.1109/ICIC.2018.8780480>.
- Seo, S., Park, S., Kim, J., 2017. Improvement of network intrusion detection accuracy by using restricted Boltzmann machine. In: *Proceedings - 2016 8th International Conference on Computational Intelligence and Communication Networks, CICN 2016*. <http://dx.doi.org/10.1109/CICN.2016.87>.
- Sethi, K., Kumar, R., Prajapati, N., Bera, P., 2020. Deep reinforcement learning based intrusion detection system for cloud infrastructure. In: *2020 International Conference on Communication Systems and Networks, COMSNETS 2020*. <http://dx.doi.org/10.1109/COMSNETS48256.2020.9027452>.
- Sethi, K., Madhav, Y.V., Kumar, R., Bera, P., 2021. Attention based multi-agent intrusion detection systems using reinforcement learning. *J. Inform. Secur. Appl.* <http://dx.doi.org/10.1016/j.jisa.2021.102923>.
- Shi, G., He, G., 2021. Collaborative multi-agent reinforcement learning for intrusion detection. In: *Proceedings of 2021 7th IEEE International Conference on Network Intelligence and Digital Content, IC-NIDC 2021*. <http://dx.doi.org/10.1109/IC-NIDC54101.2021.9660402>.
- Sigaud, O., Buffet, O., 2013. Markov Decision Processes in Artificial Intelligence: MDPs, beyond MDPs and Applications. <http://dx.doi.org/10.1002/9781118557426>.
- Sutton, R.S., Barto, A.G., 2018. *Reinforcement Learning: An Introduction, second ed.* Learning.
- Suwanalai, E., Polprasert, C., 2020. Network intrusion detection systems using adversarial reinforcement learning with deep Q-network. In: *International Conference on ICT and Knowledge Engineering*. <http://dx.doi.org/10.1109/ICTKE50349.2020.9289884>.
- Tao, J., Han, T., Li, R., 2021. Deep-reinforcement-learning-based intrusion detection in aerial computing networks. *IEEE Netw.* <http://dx.doi.org/10.1109/MNET.011.2100068>.

- Tavallae, M., Bagheri, E., Lu, W., Ghorbani, A.A., 2009. A detailed analysis of the KDD CUP 99 data set. In: IEEE Symposium on Computational Intelligence for Security and Defense Applications, CISDA 2009. <http://dx.doi.org/10.1109/CISDA.2009.5356528>.
- Tawalbeh, L., Muheidat, F., Tawalbeh, M., Quwaider, M., 2020. IoT privacy and security: Challenges and solutions. *Appl. Sci.* <http://dx.doi.org/10.3390/AP10124102>.
- Vasilyev, V.I., Lozhnikov, P.S., Sulavko, A.E., Fofanov, G.A., Zhumazhanova, S., S., S., 2018. Flexible fast learning neural networks and their application for building highly reliable biometric cryptosystems based on dynamic features. *IFAC-PapersOnLine* 51, 527–532. <http://dx.doi.org/10.1016/j.ifacol.2018.11.272>.
- Venturi, A., Apruzzese, G., Andreolini, M., Colajanni, M., Marchetti, M., 2021. DReLAB - deep reinforcement learning adversarial botnet: A benchmark dataset for adversarial attacks against botnet intrusion detection systems. *Data Brief* <http://dx.doi.org/10.1016/j.dib.2020.106631>.
- Wang, K., Chen, C.M., Hossain, M.S., Muhammad, G., Kumar, S., Kumari, S., 2021. Transfer reinforcement learning-based road object detection in next generation IoT domain. *Comput. Netw.* <http://dx.doi.org/10.1016/j.comnet.2021.108078>.
- Wang, M., Cui, Y., Wang, X., Xiao, S., Jiang, J., 2018a. ML for networking: Workflow, advances and opportunities. *IEEE Netw.* <http://dx.doi.org/10.1109/MNET.2017.1700200>.
- Wang, C., Gaimu, X., Li, C., Zou, H., Wang, W., 2019. Smart mobile crowdsensing with urban vehicles: A deep reinforcement learning perspective. *IEEE Access.* <http://dx.doi.org/10.1109/ACCESS.2019.2905263>.
- Wang, H., Han, B., Su, J., Wang, X., 2018b. A high-performance intrusion detection method based on combining supervised and unsupervised learning. In: *Proceedings - 2018 IEEE SmartWorld, Ubiquitous Intelligence and Computing, Advanced and Trusted Computing, Scalable Computing and Communications, Cloud and Big Data Computing, Internet of People and Smart City Innovations, SmartWorld/UIC/ATC/ScalCom/CBDCCom/IoP/SCI 2018*. <http://dx.doi.org/10.1109/SmartWorld.2018.00304>.
- Wang, X., Wang, C., Li, X., Leung, V.C.M., Taleb, T., 2020. Federated deep reinforcement learning for Internet of Things with decentralized cooperative edge caching. *IEEE Internet Things J.* <http://dx.doi.org/10.1109/JIOT.2020.2986803>.
- Wei, L., Zhong-Ming, Y., Ya-Ping, C., Bin, Z., 2017. A clustering algorithm oriented to intrusion detection. In: *Proceedings - 2017 IEEE International Conference on Computational Science and Engineering and IEEE/IFIP International Conference on Embedded and Ubiquitous Computing, CSE and EUC 2017*. <http://dx.doi.org/10.1109/CSE-EUC.2017.173>.
- Wu, Y., Wang, Z., Ma, Y., Leung, V.C.M., 2021. Deep reinforcement learning for blockchain in industrial IoT: A survey. *Comput. Netw.* <http://dx.doi.org/10.1016/j.comnet.2021.108004>.
- Xia, Y., Dong, S., Peng, T., Wang, T., 2021. Wireless network abnormal traffic detection method based on deep transfer reinforcement learning. In: *Proceedings - 2021 17th International Conference on Mobility, Sensing and Networking, MSN 2021*. <http://dx.doi.org/10.1109/MSN53354.2021.00083>.
- Xiao, L., Jiang, D., Xu, D., Su, W., An, N., Wang, D., 2018a. Secure mobile crowdsensing based on deep learning. *China Commun.* <http://dx.doi.org/10.1109/CC.2018.8485464>.
- Xiao, L., Li, Y., Han, G., Liu, G., Zhuang, W., 2016. PHY-layer spoofing detection with reinforcement learning in wireless networks. *IEEE Trans. Veh. Technol.* <http://dx.doi.org/10.1109/TVT.2016.2524258>.
- Xiao, L., Li, Y., Huang, X., Du, X., 2017. Cloud-based malware detection game for mobile devices with offloading. *IEEE Trans. Mob. Comput.* <http://dx.doi.org/10.1109/TMC.2017.2687918>.
- Xiao, L., Li, Y., Liu, G., Li, Q., Zhuang, W., 2015. Spoofing detection with reinforcement learning in wireless networks. In: *2015 IEEE Global Communications Conference. GLOBECOM 2015*, <http://dx.doi.org/10.1109/GLOCOM.2014.7417078>.
- Xiao, L., Lu, X., Xu, T., Zhuang, W., Dai, H., 2021. Reinforcement learning-based physical-layer authentication for controller area networks. *IEEE Trans. Inf. Forensics Secur.* <http://dx.doi.org/10.1109/TIFS.2021.3056206>.
- Xiao, L., Sheng, G., Wan, X., Su, W., Cheng, P., 2019. Learning-based PHY-layer authentication for underwater sensor networks. *IEEE Commun. Lett.* <http://dx.doi.org/10.1109/LCOMM.2018.2877317>.
- Xiao, L., Wan, X., Dai, C., Du, X., Chen, X., Guizani, M., 2018b. Security in mobile edge caching with reinforcement learning. *IEEE Wirel. Commun.* <http://dx.doi.org/10.1109/MWC.2018.1700291>.
- Xu, T., Lu, X., Xiao, L., Tang, Y., Dai, H., 2019. Voltage based authentication for controller area networks with reinforcement learning. In: *IEEE International Conference on Communications*. <http://dx.doi.org/10.1109/ICC.2019.8761744>.
- Xu, H., Qiu, X., Zhang, W., Liu, K., Liu, S., Chen, W., 2021. Privacy-preserving incentive mechanism for multi-leader multi-follower IoT-edge computing market: A reinforcement learning approach. *J. Syst. Archit.* <http://dx.doi.org/10.1016/j.sysarc.2020.101932>.
- Xue, Z., Zhou, P., Xu, Z., Wang, X., Xie, Y., Ding, X., Wen, S., 2021. A resource-constrained and privacy-preserving edge-computing-enabled clinical decision system: A federated reinforcement learning approach. *IEEE Internet Things J.* <http://dx.doi.org/10.1109/JIOT.2021.3057653>.
- Yang, L., Yao, H., Wang, J., Jiang, C., Benslimane, A., Liu, Y., 2020. Multi-UAV-enabled load-balance mobile-edge computing for IoT networks. *IEEE Internet Things J.* <http://dx.doi.org/10.1109/JIOT.2020.2971645>.
- Zhan, Y., Li, P., Qu, Z., Zeng, D., Guo, S., 2020. A learning-based incentive mechanism for federated learning. *IEEE Internet Things J.* <http://dx.doi.org/10.1109/JIOT.2020.2967772>.
- Zhu, L., Liang, H., Wang, H., Ning, B., Tang, T., 2021a. Joint security and train control design in blockchain empowered CBTC system. *IEEE Internet Things J.* <http://dx.doi.org/10.1109/JIOT.2021.3097156>.
- Zhu, P., Zhang, J., Xiao, Y., Cui, J., Bai, L., Ji, Y., 2021b. Deep reinforcement learning-based radio function deployment for secure and resource-efficient NG-RAN slicing. *Eng. Appl. Artif. Intell.* <http://dx.doi.org/10.1016/j.engappai.2021.104490>.